

Two-Phase Search for Optimal Random Forest Tree Count

Seven Theorems Proving $O(\log N)$ Optimality

Pre-submission draft · Revised April 2026

Note on numerical results: All values in Tables 1–7 and Appendix A are theoretical projections derived from the closed-form complexity bounds in Section 6 and order-of-magnitude estimates from prior literature. Empirical values will replace these before submission.

Overview

This repository contains the paper, proofs, and experimental code for the **Two-Phase Search** algorithm — a provably optimal method for selecting the minimum sufficient tree count k^* in Random Forest models.

The algorithm exploits the monotonic-then-plateau structure of the RF accuracy curve to find k^* in $O(\log N)$ function evaluations, matching the information-theoretic lower bound (Theorem 7). No comparison-based algorithm can do better in the worst case.

The Algorithm

Two-Phase Search runs in two stages:

- Phase 1 — Exponential Bracketing.** Starting from $k_{\min}$, double the tree count at each step ($k_i = k_{\min} \cdot 2^i$) until the plateau detection criterion is satisfied. This brackets k^* in at most $\lceil \log_2(k^*/k_{\min}) \rceil + 1$ evaluations.
- Phase 2 — Binary Refinement.** Apply binary search over the discrete bracket $[L, U]$ from Phase 1 to localise k^* to within ϵ -optimality in $\lceil \log_2(U - L) \rceil$ additional evaluations.

Total cost: $O(\log N + \log(L/\epsilon))$ — provably optimal by Theorem 7.

Theoretical Results (Seven Theorems)

#	Theorem	Result
1	Phase 1 Complexity	Exponential doubling brackets k^* in $O(\log N)$ evaluations

#	Theorem	Result
2	Plateau Detection	Windowed gradient criterion correct with $P(\text{error}) \leq \exp(-w\varepsilon^2/8\sigma^2)$
3	Phase 2 Complexity	Discrete binary search terminates in exactly $\lceil \log_2(M) \rceil$ steps
4	Combined ε -Optimality	Algorithm finds ε -optimal k^* in $O(\log N)$ total evaluations
5	Convergence Rate	Position error decays geometrically: $ k_t - k^* \leq (U-L)/2^t$
6	Noise Robustness	False positive rate $\leq 2 \cdot \exp(-w\varepsilon^2/2\sigma^2)$ under bounded noise
7	Information-Theoretic Lower Bound	$\Omega(\log N + \log(L/\varepsilon))$ — Algorithm 1 is asymptotically optimal

All proofs operate on the discrete integer domain under four formally stated assumptions (monotonicity, plateau threshold, discrete Lipschitz continuity, bounded noise).

Why This Matters

The standard practitioner heuristic — train with 500 trees and inspect the OOB error curve — is statistically sound for interactive desktop or cloud workflows. This algorithm addresses the contexts where that approach breaks down:

- **Edge / embedded deployment** — hard memory and latency budgets; OOB monitoring unavailable post-deployment.
- **AutoML pipelines** — the $O(\log N)$ vs $O(N)$ reduction compounds across thousands of configurations.
- **Production systems with latency SLAs** — the ε -optimality guarantee enables principled tree count reduction.
- **Research benchmarking** — removes an implicit confound from comparative evaluations.

Projected results on Coverttype, MNIST, and Adult suggest **7–15× speedup** over grid search with solutions within 0.01 accuracy in over 95% of runs. Two-Phase Search is, to our knowledge, the only sub-grid-search method offering a **provable ε -optimality guarantee** on this problem.

Repository Structure



```

├─ paper/
│   └─ Two_Phase_Search_Paper_Clean_v4.docx    # Current draft
├─ algorithm/
│   └─ two_phase_search.py                    # Algorithm 1 implementation
(pending)
├─ experiments/
│   └─ baselines.py                          # Grid search, random search,
Bayesian opt
│   └─ run_experiments.py                    # Tables 1-7 and Appendix A
│   └─ figures/                             # Figure generation scripts
└─ README.md

```

The `algorithm/` and `experiments/` directories are placeholders pending the empirical validation runs described in Section 7 of the paper.

Experimental Setup

- **Datasets:** Covertypes, MNIST, Adult
- **Baselines:** Grid search (N=100), Random search (50 evals), Bayesian optimisation (30 iters, GP + Matérn kernel)
- **Configuration:** $k_{\min}=10$, $k_{\max}=1000$, $\epsilon=0.001$, $w=5$ (recommended window size)
- **Runs:** 10 per configuration, seeds 42-51; reported as mean $\pm$ std
- **Environment:** scikit-learn 1.2.0, Python 3.8+

Status

Component	Status
Theoretical proofs (7 theorems)	✓ Complete
Algorithm pseudocode (Algorithm 1)	✓ Complete
Related work (Section 3)	✓ Complete
Pre-publication review revisions	✓ All resolved (see Appendix B)
Empirical validation (Tables 1-7)	⌚ Pending
Figure generation (Figures 0-3)	⌚ Pending

Component	Status
FPGA / microcontroller validation	☐ Future work

Planned Submission

A hyperparameter optimisation workshop (AutoML, NeurIPS HPO Workshop), with a path toward a top-tier venue once empirical results are in hand.

References

Key prior work this builds on:

- Oshiro et al. (2012) — tree count convergence across 29 datasets
- Probst & Boulesteix (2017) — systematic analysis across 99 datasets (empirical grounding for Assumption 1)
- Pham-Quoc et al. (2024) — FPGA-based RF acceleration; memory as primary deployment constraint
- Daghero et al. (2022) — RF inference on RISC-V microcontrollers; energy cost analysis
- Bergstra & Bengio (2012), Snoek et al. (2012) — random search and Bayesian optimisation baselines